

PATIENT HEALTH TRACKING SYSTEM (PHTS)

Semester Project Report

Course: Information & Communication Theory

Program: BS Data Science (Semester 1)

Project Type: Theory

Group Members:

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Executive Summary:

Healthcare systems generate large volumes of patient vital data, but raw data alone is insufficient for timely clinical decisions. This project proposes a Patient Vital Monitoring and Risk Alert System that collects patient vitals, cleans and aggregates the data, stores it in structured datasets, and generates alerts to assist healthcare professionals.

The system integrates C++ for data input and alert generation, Python for data cleaning and preprocessing, and Microsoft Access as a relational database for structured storage. Statistical features such as mean, minimum, maximum, and derived risk indicators are computed to identify patient deterioration risks.

The final outcome is a semi-automated decision support pipeline that improves data quality, clinical visibility, and response time for patient monitoring.

Problem Statement:

In many healthcare environments, patient vital data is either:

- stored in unstructured formats,
- manually analyzed,
- or not analyzed in real-time.

This leads to:

- delayed clinical decisions,
- missed early warning signs,
- inefficient use of healthcare resources.

There is a need for a lightweight, data-driven system that:

- standardizes patient vital data,
- performs basic risk analysis,
- and provides clear alerts for clinicians.

Objectives:

The main objectives of this project are:

1. To design a structured pipeline for patient vital data collection
2. To clean and aggregate raw vital data into meaningful patient-level features
3. To generate automated alerts based on risk indicators

4. To store data in both CSV and relational database formats
5. To provide a reusable and scalable system architecture

Scope of the Project:

Included

- Patient vital data input (heart rate, SpO₂, BP, temperature, etc.)
- Data cleaning and aggregation
- Risk indicator calculation
- Alert generation
- CSV-based dataset management
- Microsoft Access database design

Excluded

- Real-time sensor integration
- Machine learning prediction models
- Hospital-level deployment

Stakeholders:

Stakeholder	Interest
Doctors	Quick risk assignment
Nurses	Monitoring patient vitals
Hospital Admin	Data organization
Students / Developers	Learning data pipelines

Comparative Study & Gap Analysis

Existing Systems

- Hospital Monitoring Systems
- ICU Dashboards
- Wearable Health Trackers

Limitations

- Expensive infrastructure
- Complex hardware dependencies
- Limited data transparency

Identified Gap

Most systems do not provide:

- simple CSV-based integration,
- transparent feature engineering,
- student-friendly implementation.

Proposed Solution

A lightweight system using:

- C++ for logic and alerts
- Python for data cleaning
- Access for structured storage

Methodology

1. Data Collection

- User inputs patient ID, age, gender
- Hour-wise vitals are entered
- Raw data stored in a separate CSV

2. Data Cleaning (Python)

- Handling missing values
- Converting data types to float

- Removing invalid readings
- Calculating summary statistics

3. Feature Engineering

- Mean, min, max of vitals
- Derived indicators (ranges, flags)
- Risk score calculation

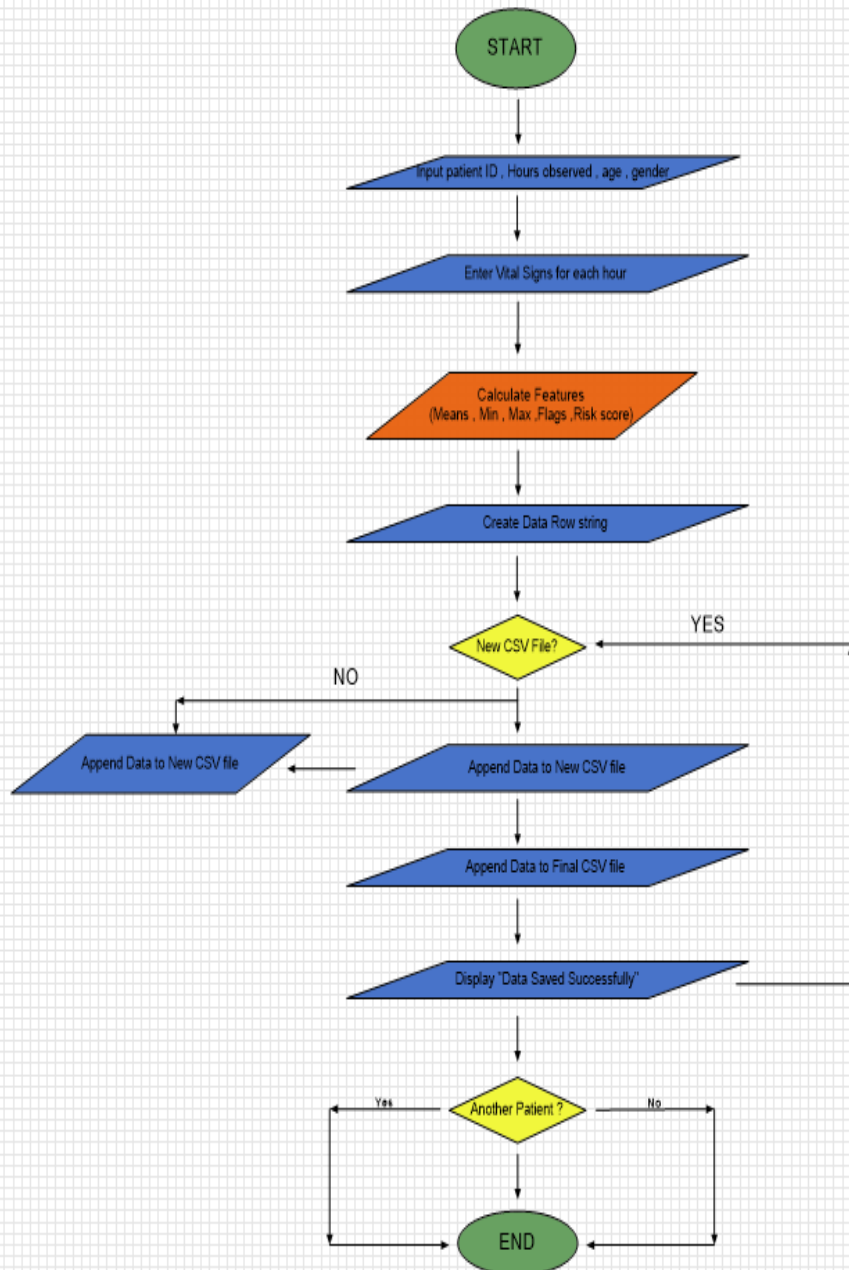
4. Integration (C++)

- Cleaned data appended to final dataset
- Alerts generated by scanning CSV
- User-based patient search supported

5. Database Storage (Access)

- Structured relational tables
- Patient-level and vitals-level separation

Data Pipeline Overview:



System Architecture

- **Frontend (CLI):** C++
- **Processing Layer:** Python
- **Storage Layer:** CSV + Access DB
- **Output:** Alerts & summaries

Functional Requirements

- Input patient data
- Store raw and cleaned data
- Append new records
- Search patient by ID
- Generate alerts
- Support multiple patient entries

Non-Functional Requirements

- Data accuracy
- File integrity
- Usability
- Maintainability
- Portability

Database Schema (Access)

Patient Table

- patient_id (PK)
- age
- gender

Vitals Table

- vital_id (PK)
- patient_id (FK)
- heart_rate
- spo2
- bp
- temperature
- timestamp

Risk Table

- patient_id
- risk_score
- alert_flags

Ethical Considerations

- Patient anonymity maintained
- No real patient identifiers
- Educational use only
- No clinical diagnosis claims

Limitations & Future Work

Limitations

- Rule-based alerts
- No ML prediction
- Manual data input

Future Enhancements

- Machine learning risk prediction
- Real-time sensor integration
- Web-based dashboard
- Hospital system integration

Conclusion

This project demonstrates how healthcare data can be transformed into actionable insights using a structured data pipeline. By integrating C++, Python, and Access Database, the system achieves efficient data handling, risk analysis, and alert generation, making it a valuable educational and practical prototype.

References

1. WHO – Patient Monitoring Guidelines
2. UCI Machine Learning Repository
3. CDC – Vital Signs Data
4. IEEE Healthcare Analytics Paper

Workload Distribution Among Team

Members:

Member	Role	Assigned Responsibilities
Abdul Ahad	Team Lead / Core Developer	Overall project coordination, complete C++ core logic, alert generation module, CSV integration, data pipeline implementation, Python-based data cleaning & preprocessing

Mehreen	Data & Finance Analyst	Financial forecasting (cost estimation, resource planning), RAPTOR flowchart design, contribution to feasibility analysis
Mahnoor	UX / Database Designer	Figma UI wireframes, Microsoft Access database schema design, ER diagrams, table relationships, basic UI documentation
Maryam	Documentation & Research Support	Literature review, comparative study support, gap analysis writing, formatting & proofreading of final theory document, reference management (APA style)